# Solution: 'K' Closest Points to the Origin

## Problem Statement

Given an array of points in a 2D plane, find 'K' closest points to the origin.

**Example 1:**

```
Input: points = [[1,2],[1,3]], K = 1
Output: [[1,2]]
Explanation: The Euclidean distance between (1, 2) and the origin is sqrt(5).
The Euclidean distance between (1, 3) and the origin is sqrt(10).
Since sqrt(5) < sqrt(10), therefore (1, 2) is closer to the origin.
```

**Example 2:**

```
Input: point = [[1, 3], [3, 4], [2, -1]], K = 2
Output: [[1, 3], [2, -1]]
```

## Solution

The `Euclidean distance` of a point P(x,y) from the origin can be calculated through the following formula:

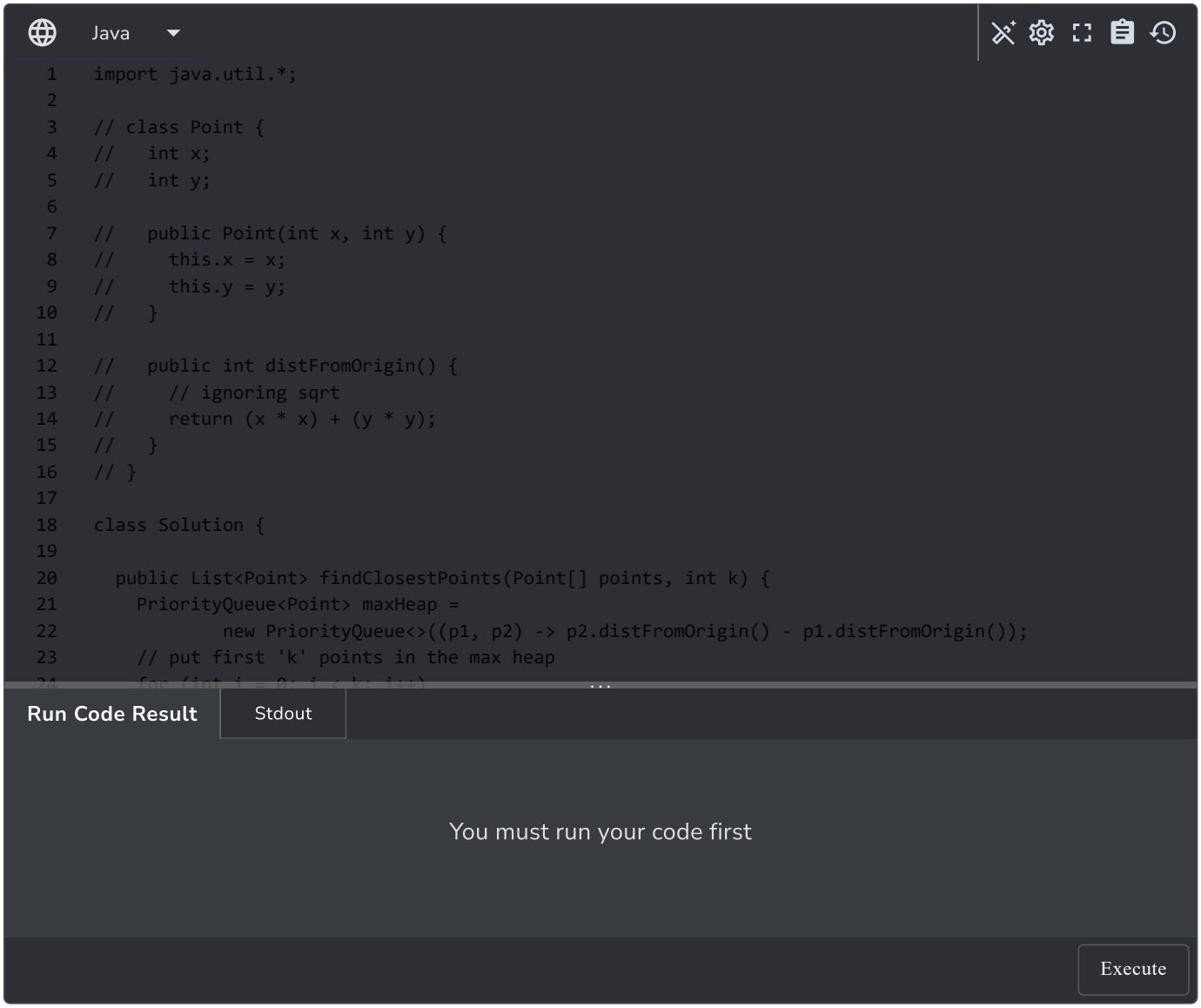
$$\sqrt{x^2 + y^2}$$

This problem follows the `Top 'K' Numbers pattern`. The only difference in this problem is that we need to find the closest point (to the origin) as compared to finding the largest numbers.

Following a similar approach, we can use a **Max Heap** to find 'K' points closest to the origin. While iterating through all points, if a point (say 'P') is closer to the origin than the top point of the max-heap, we will remove that top point from the heap and add 'P' to always keep the closest points in the heap.

## Code

Here is what our algorithm will look like:




## Time Complexity

The time complexity of this algorithm is $O(N * logK)$ as we iterating all points and pushing them into the heap.

## Space Complexity

The space complexity will be $O(K)$ because we need to store 'K' point in the heap.


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